SST Falsifier Report

A063-v0.1.0
Hicks-Hill analytic Euler separatrix controls
External read-only rendering; not an original frozen report.
Generated: 2026-10-13T10:29:19.945378+00:00

Evidence boundary

No new science was executed for this report. Gate outcomes are copied from the source ledger without reclassification. Missing records are not silently filled.

Gate summary

PASS=9, UNRESOLVED=2

Gate	Status	Preregistered question
H0	PASS	Protocol/source integrity recorded?
H1	PASS	Do analytical roots satisfy registered equations?
H2	PASS	Does Hill zero-swirl control satisfy Euler/divergence?
H3	PASS	Does first Hicks Beltrami field satisfy curl(u)=k*u and stationary Euler?
H4	PASS	Does Hicks lambda2 first mode have zero translation?
H5	PASS	Is the spherical boundary a material surface in the co-moving frame?
H6	PASS	Do internal/external pressure and velocity match at the surface?
H7	UNRESOLVED	Are torus-knot accessibility and finite-core vortex-tube stability certified?
H8	UNRESOLVED	Has independent C++ FP64 certification run through v1.0.7 frozen protocol?
H9	PASS	Does exact pressure satisfy incompressible pressure-Poisson closure?
H10	PASS	Does finite laboratory-frame kinetic-energy quadrature converge?

Gate detail (source-derived)

H0 — PASS

Protocol/source integrity recorded?

Not a v1.0.7 frozen framework instance.

- protocol_sha256: 7e002607b71dbf520761d0072916a742c00bc3782bef7eac9a0bf1e7d131f498
- scope: standalone analytical prototype

H1 — PASS

Do analytical roots satisfy registered equations?

Hicks lambda1/lambda2 control equations.

- max_root_equation_abs_residual: 9.547918011776346e-14
- lambda1: array, 3 elements
- lambda2: array, 3 elements

H2 — PASS

Does Hill zero-swirl control satisfy Euler/divergence?

Analytic Hill control: not proof of stability.

- max_hill_euler_residual: 3.684793422721659e-08

H3 — PASS

Does first Hicks Beltrami field satisfy curl(u)=k*u and stationary Euler?

Finite differences at points strictly inside sphere; Python reference only.

- max_beltrami: 6.820200743323078e-07
- max_euler: 1.7049375007482077e-05
- max_div: 4.3346847355962836e-07

H4 — PASS

Does Hicks lambda2 first mode have zero translation?

Zero external aggregate translation does not imply zero internal motion.

- translation_U: 5.921189464667501e-16

H5 — PASS

Is the spherical boundary a material surface in the co-moving frame?

No normal flux across spherical separatrix in exact field.

- max_surface_normal_velocity: 4.440892098500626e-16
- max_surface_psi: 0.0

H6 — PASS

Do internal/external pressure and velocity match at the surface?

One matched mid-latitude point, finite-offset diagnostic only.

- boundary_velocity_jump_eps1e7: 1.9092742863117333e-06
- pressure_jump_eps1e7: 1.3851301678968753e-06

H7 — UNRESOLVED

Are torus-knot accessibility and finite-core vortex-tube stability certified?

Five closed vortex-line winding diagnostics do NOT classify finite vortex tubes or certify inaccessible topologies.

- sampled_windings: array, 5 elements

H8 — UNRESOLVED

Has independent C++ FP64 certification run through v1.0.7 frozen protocol?

Must instantiate official v1.0.7 template and run native certification; not executed in this portable prototype.

- backend: Python FP64 reference only

H9 — PASS

Does exact pressure satisfy incompressible pressure-Poisson closure?

A046 analytical field control at two interior samples; no gravity metric claims.

- max_pressure_poisson_abs: 0.000279955396990772

H10 — PASS

Does finite laboratory-frame kinetic-energy quadrature converge?

A057 reference energy control only; not at matched fixed circulation or topology and not a mass ratio.

- max_order36_60_relative_error: 6.731390939299886e-15

Backend provenance

{"cpp_certification": {"status": "NOT_RUN"}, "framework_target": "v1.0.7; not instantiated", "python_reference": {"authority": "REFERENCE", "precision": "FP64"}, "schema": "SST-HICKS-PROTOTYPE-BACKENDS"}

Report provenance

Declared ledger-to-summary digest binding: True
Files read: BACKEND_MANIFEST.json, GATE_LEDGER.json, RUN_SUMMARY.json
Scope: Read-only digest inventory and declared ledger/summary binding check only. Does not independently certify framework protocol, scientific results, or frozen output tree.